

Defining a Platform Approach and Market Participation: Data Driven Business Models for Solid State Transformer-Based Synthetic Inertia and Voltage Stability Controls

Cooperative Research and Development Final Report

CRADA Number: CRD-21-21469, Project 1, Mod 1

NREL Technical Contacts: Harsha Vardhana Padullaparti, Priti Paudyal, and Tom Harris

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Cooperative Research and Development Final Report

Report Date: July 9, 2025

In accordance with requirements set forth in the terms of the CRADA agreement, this document is the CRADA report, including a list of subject inventions, to be forwarded to the DOE Office of Scientific and Technical Information as part of the commitment to the public to demonstrate results of federally funded research.

Parties to the Agreement: Eaton Corporation

CRADA Number: CRD-21-21469, Project 1, Mod 1

Project 1 Title: Defining a Platform Approach and Market Participation: Data Driven Business Models for Solid State Transformer-Based Synthetic Inertia and Voltage Stability Controls

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Sponsoring DOE Program Office(s):

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Joint Work Statement Funding Table showing DOE Commitment:

No shared resources

Estimated Costs	NREL Shared Resources Government In-Kind
Year 1 Year 2, Modification #1	$0.00 $0.00
TOTALS	$0.00

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Executive Summary of CRADA Work:

The primary objective of this project is to determine the incremental value created with the medium voltage solid-state transformer (MV SST) technology to different stakeholders in view of the updated DER grid regulations. This includes studying the benefits of the MV SST technology in a range of use cases for EV and DER penetration including (1) "corridor charging" for EVs and (2) solar plus storage (FERC 2222). The potential customers of this technology include utilities for EV charging, DER installers who must meet utility interconnection requirements, balancing authorities, and DER aggregators. The traditional transformers on the grid could be a limiting factor for the EV-grid integration as the distribution transformers were not designed to handle the dynamic and fluctuating EV charging loads. Thus, the issues such as voltage fluctuations, increased losses and reduced efficiency [1] can negatively impact the grid operation. To address these challenges, transformers with flexibility and adaptability become imperative to meet the evolving energy demands. In this regard, the concept of Medium Voltage Solid-State Transformers.

CRADA benefit to DOE, Participant, and US Taxpayer:

- Assists laboratory in achieving programmatic scope.

(MV SST) is a potential solution. MV SSTs are advanced power electronic devices capable of regulating voltage levels and efficiently managing power flow in the distribution system. Studies have shown that SSTs can effectively enhance grid stability, provide faster response times, and reduce losses compared to traditional transformers [2]. The potential applications for MV SSTs include integration of EVs [3], and grid integration of DERs such as PVs and batteries [3-10], and electric aircraft [11]. By incorporating SSTs into the distribution system, grid operators can enhance the grid's capacity to handle the influx of EVs and maintain stable voltage levels.

Summary of Research Results:

Task 1 Develop feeder and solid state transformer models: In this task, the utility distribution feeders will be selected in consultation with utility engineers, as available. The selected feeder models will be converted to a distribution power flow solution tool by NREL team to perform the analysis in the subsequent tasks. The SST model is not available in the distribution network simulators to study the effectiveness and the value of SST for a distribution system of the future which incorporates multiple SSTs. Therefore, NREL team will develop a model of the SST useful for power flow calculations. This model will be used by NREL team to emulate the SST functionality in terms of real and reactive power injections as well as to record the necessary data from the power flow simulations for the impact and value analysis. Estimated completion: 3 months from project start.

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Distribution system feeder model:

Actual distribution system feeder models are not publicly available to conduct research studies. There are test feeders, designed to replicate the behavior of actual distribution feeders. Several publicly available test feeders were considered for the selection of an appropriate feeder for this study. Table 1 shows the different test feeders considered. Among these feeders, EPRI J1 feeder was selected as per the project team's discussion. The J1 feeder contains a greater number of voltage regulators and capacitors. In this project, we studied the application of medium voltage solid-state transformers, its value and impact in coordination with the voltage regulators, so the EPRI J1 feeder best suited the selection criteria. Figure 1 shows the J1 feeder topology. This feeder is in the northeastern US and consists of around 1300 customers including residential, commercial, and light industrial customers. It has 58 circuit miles of primary network. Almost 30% of PV penetration exists in this feeder model.

Table 1. Test Feeder Selection

Feeder	Length (km)	Primary Voltage (kV)	Number of customers/Loads	Number of devices		
				Voltage regulator	Capacitor bank	LTC
IEEE 13 node	2.5	4.16	9	1	2	
IEEE 123 node	12	4.16	114	3	4	1
IEEE 34 node	94	24.9	24	2	2	
EPRI J1	93.3	12.47	1384	8	5	1
EPRI K1	45.1	13.2	321		1	1
EPRI M1	20.9	12.47	1470		3	1

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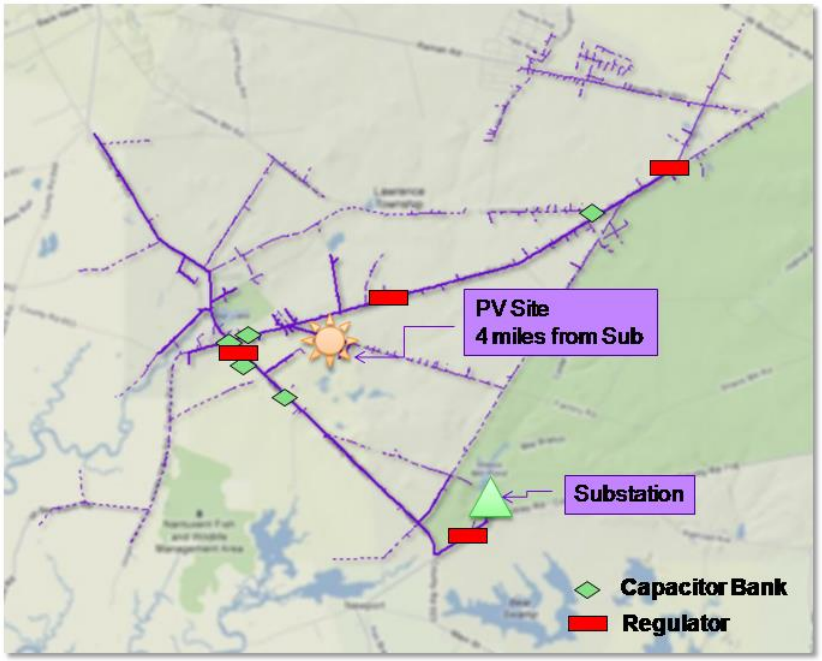


Figure 1. EPRI J1 feeder topology.

Solid state transformer model:

A solid-state transformer is a power electronic transformer, an AC-to-AC converter. It enables bi-directional power flow and can enhance power quality performance. It can support different smart grid challenges over conventional transformers such as voltage sag compensation, instantaneous voltage regulation, harmonic compensation, power factor correction, auto-balancing, short circuit protection, and variable-frequency output [1]. The SST model in OpenDSS is represented as two different components. A medium voltage load on the MV (primary) side and a low voltage source on the LV (secondary) side as shown in Figure 2. The LV side supplies the power required by the loads on the LV side and maintains a constant voltage at the secondary terminal. As SST supports bidirectional power flow, this SST model can operate in 2 modes: Load mode and Generator mode. The load on the MV side resembles the SST behavior. The active power on the MV load is equal to the active power supplied by the LV source plus the SST losses. It equals the active power injected from the LV terminal minus the SST losses, in case of reverse power flow. SST efficiency is considered as 98% in this study.

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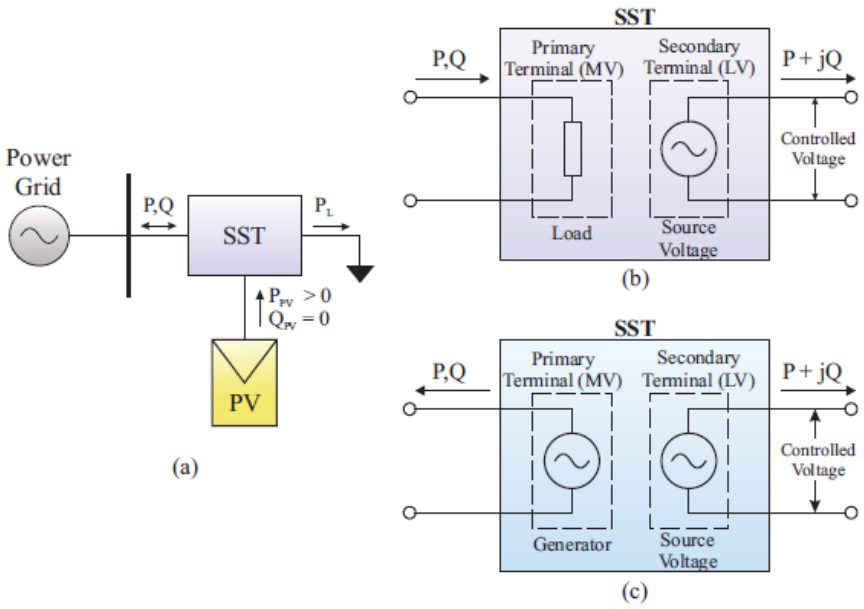


Figure 2. Solid state transformer model in OpenDSS. (a) A basic circuit with SST, (b) SST model in load mode of operation, (c) SST model in generator mode of operation [12].

Input

- Rated capacity of the transformer
- Rated Voltage levels
- SST efficiency

Features

- Bidirectional operation
- Controlled voltage on the LV side
- Limits for primary voltage (MV side); maximum and minimum voltage limit, SST turnoff
- Active and reactive power flow limit defined similar as Fig 3.

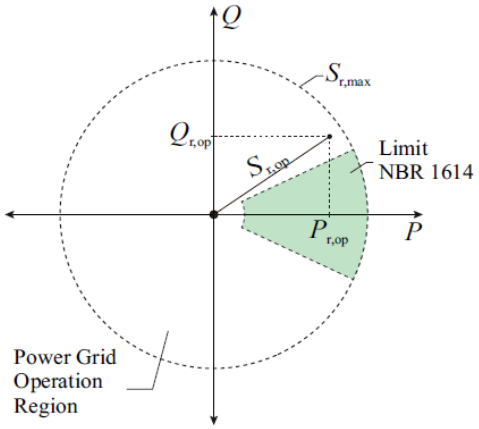


Figure 3. PQ operation.

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The two components, MV side load and LV side voltage source, are defined in OpenDSS with some random load value at first. Then, the actual MV side load value for every timestamp is adjusted through Python. The following steps are executed for the power flow simulation with SST using Python [1]:

1. First, solve the system snapshot with a random value for the SST load.
2. Measure the total power supplied by the LV side voltage source.
3. Calculate the power demand on the MV network by adding the SST losses.
4. Update the MV side load kW with the power calculated in 3.
5. Solve the system with updated values.
6. Check MV side voltage if there is any voltage violation.
7. If yes, then increase/decrease the reactive power on the MV side load by a fixed step-size to perform reactive power control. Check and keep on updating the reactive power until the violation is mitigated or until the SST limit is reached. Get the final solution.
8. If not, then move to another timestep.

This process is repeated every time step power flow solution.

SST reactive power control

Reactive power control functionality at the MV terminal is enabled through either one of the two criteria.

1. Volt/var approach – enables reactive power support when the local voltage is beyond the defined voltage range.

2. Relative voltage sensitivity – compares present voltage to previous timestep voltage and absorbs/injects reactive power in case the difference is greater than given threshold.

An incremental approach with fixed step change of reactive power is considered to determine the reactive power injection/absorption.

$$Q_{SST} = Q_{SST} \pm \Delta Q$$

- Reactive power limit is determined by the capacity of the SST.

$$Q_{SST} = \sqrt{S_{SST}^2 - P_{SST}^2}$$

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Task 2 Develop use case scenarios: In this task, NREL team will generate a set of EV penetration scenarios to simulate the EV charging behavior and estimate charging load profiles and charging requirements. These scenarios will be realistic considering the current and future EV penetration trends, EV infrastructure requirements, and traffic patterns. Further, the increasing penetrations of other DERs such as PV introduce generation variabilities, leading to various operational issues specified above which can be mitigated through the effective use of BESS. High (50-80%) DER penetration scenarios including PV and BESS will be developed by NREL team in this task to study the system level impacts of high DER. The metrics useful for quantifying the value for a distribution system employing the SSTs will be developed by NREL team. The NREL team will refine the use case scenarios and performance metrics with Eaton team and the utility, as available, to ensure that they capture boundary conditions of operations. Estimated completion: 6 months from project start.

The focus of the work is to analyze the value of MV SST on the distribution feeder in the presence of high EV penetration. Furthermore, the study also considers the addition of PV and energy storage to support high EV charging on the grid. Sites for large EV charging capacity are considered in this study. The EV and PV penetration considered in the work are defined here.

EV penetration: The percentage of maximum coincident EV charging power over peak feeder load.

$$\left(EV_{penetration} = \frac{Max\ coincident\ EV\ chargingpower}{Feeder\ peak\ demand}\right) * 100\%$$

PV penetration: The percentage of total PV installed capacity over peak feeder load.

$$\left(PV_{penetration} = \frac{Total\ PV\ capacity}{Feeder\ peak\ demand}\right) * 100\%$$

The study considers 2 specific use case scenarios enabling high EV charging on the feeder. Comparative study of the two scenarios is performed to explore the benefits of SST on the grid.

Use case scenario 1: Scenario with EV

Several EV charging sites are selected at certain distances across the feeder to enable high power EV charging stations. In this scenario, the EV charging loads are added at specific sites on the feeder. Time-series power flow simulation is conducted for a day simulating the maximum coincident EV charging to investigate the impact of high EV charging demand on the feeder. Different power system parameters are measured, and the performance of the grid operation is evaluated through a set of metrics.

Use case scenario 2: Scenario with EV and MV SST

In this scenario, the conventional transformers at the EV charging sites are replaced by MV SSTs. Time-series power flow study with the same EV charging demand as scenario 1 is conducted, and the performance metrics are evaluated.

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Use Case Analysis with J1 Feeder:

The use cases implementation on the J1 feeder starts with the selection of suitable EV charging sites. Fig. 4 presents the selected five EV charging sites (A, B, C, D, and E) on the J1 feeder. The 3 phase nodes, voltage level, and the distance between the sites are considered in the site selection. The sites consist of 150 kW level EV chargers; Sites A and B include 12 charging ports each while sites C, D, and E have 10 ports each. Figure 5 shows the aggregated EV charging profiles at each site, and Fig. 6 represents the total aggregated EV charging on the feeder. EV penetration is 50% in this case. PV penetration is about 30% for the initial case study.

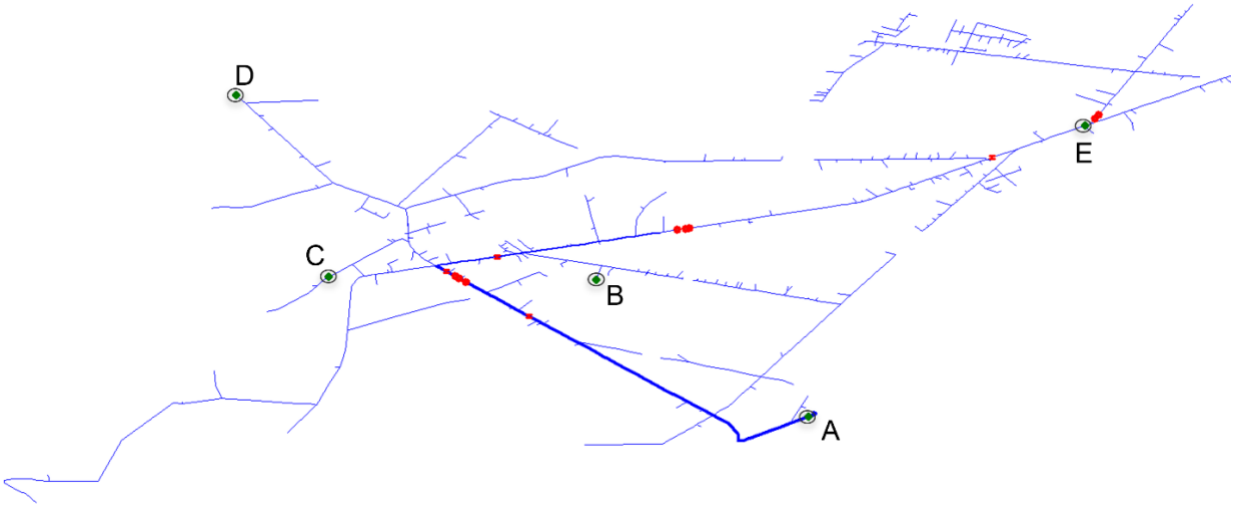


Figure 4. J1 feeder network with five EV charging sites A through E.

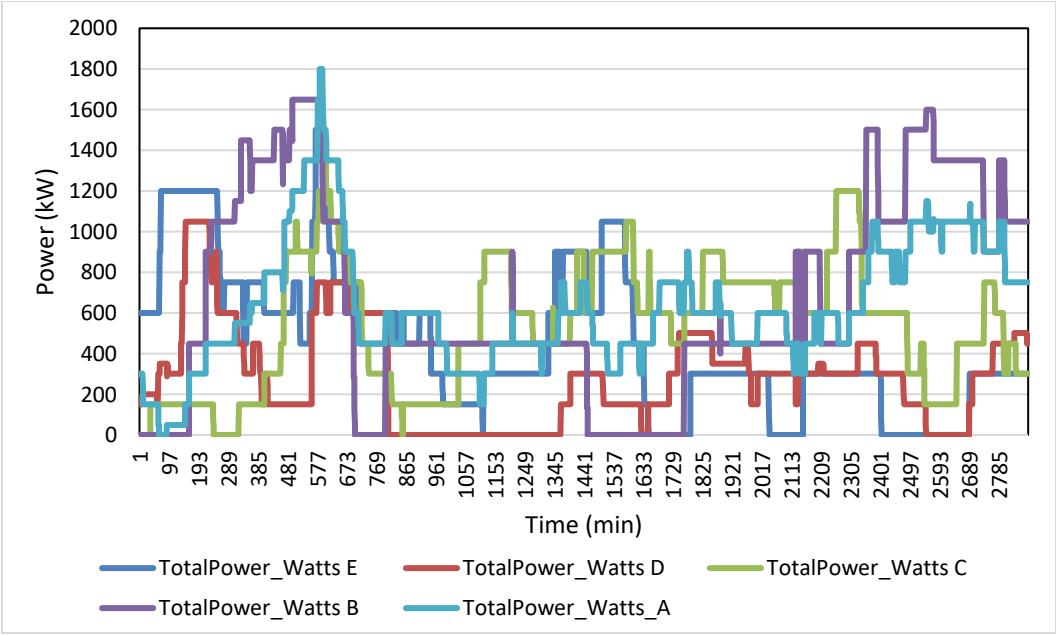


Figure 5. Individual aggregated EV charging profiles at the five sites.

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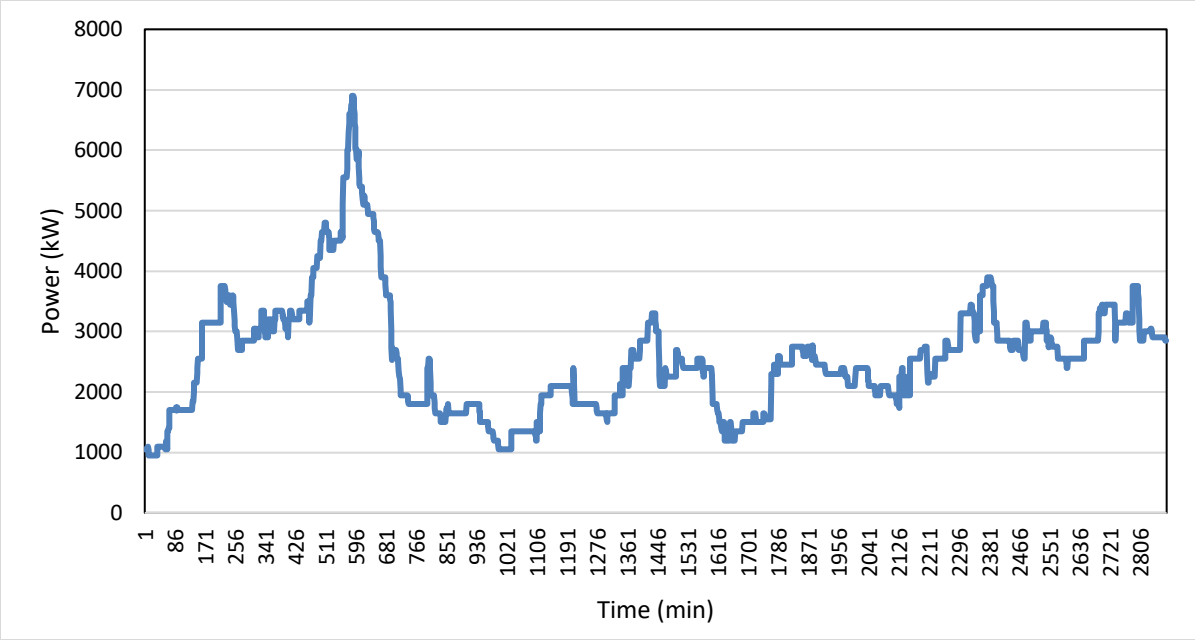


Figure 6. Total aggregated EV charging profile on the feeder.

Table 2 presents the performance metrics defined in this study. Different power system parameters such as current, voltage, and power are measured and analyzed at different levels on the feeder to determine the grid performance under the developed scenarios.

Table 2. Metrics

	Performance evaluation metrices	
	Timeseries/plots	**Values**
MV SST	kW and kVar Voltage mag. and angle (MV and LV terminals)	Overall max loading/capacity
LTC and Regulators	Individual tap position	Total number of tap change
Capacitors	ON/OFF position	ON/OFF frequency
Substation level	Total kW and kVar PV generation, kW	Peak load, kW
Feeder overall voltage	Min and max per unit	Min and max per unit Number of nodes with voltages beyond the range [0.95, 1.05] Node hours/minutes of over or under voltage
Line capacity	Histogram of percentage line loadings	Number of overloaded lines Hours/minutes of overloading Worst case line loading heatmap
EV site	Charging load profile, kW	Peak EV charging load, kW

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Later, considering the use case scenario results, an economic analysis was performed to explore the cost-benefit of a MV SST with DC-to-DC converters as against a conventional distribution transformer with downstream AC-to-DC rectifier and DC-to-DC converters.

Preliminary Analysis:

A 1-minute timestep simulation was conducted.

Voltage:

The overall minimum and maximum voltages on the feeder through the simulation day is shown in Fig. 7. In scenario 1 with EV only, it was observed that even with the active regulators and capacitors' controls, the minimum voltage goes beyond the ANSI limit of 0.95 per unit in the presence of high EV charging load. It's clear that the low voltage beyond 0.95 occurred at the time of peak EV charging. On the other hand, in scenario 2, when SST was added instead of conventional transformers, the minimum voltages were maintained above 0.95 per unit. The minimum and maximum per unit voltages, along with the number of nodes that experienced voltage beyond the ANSI range is shown in Table 3.

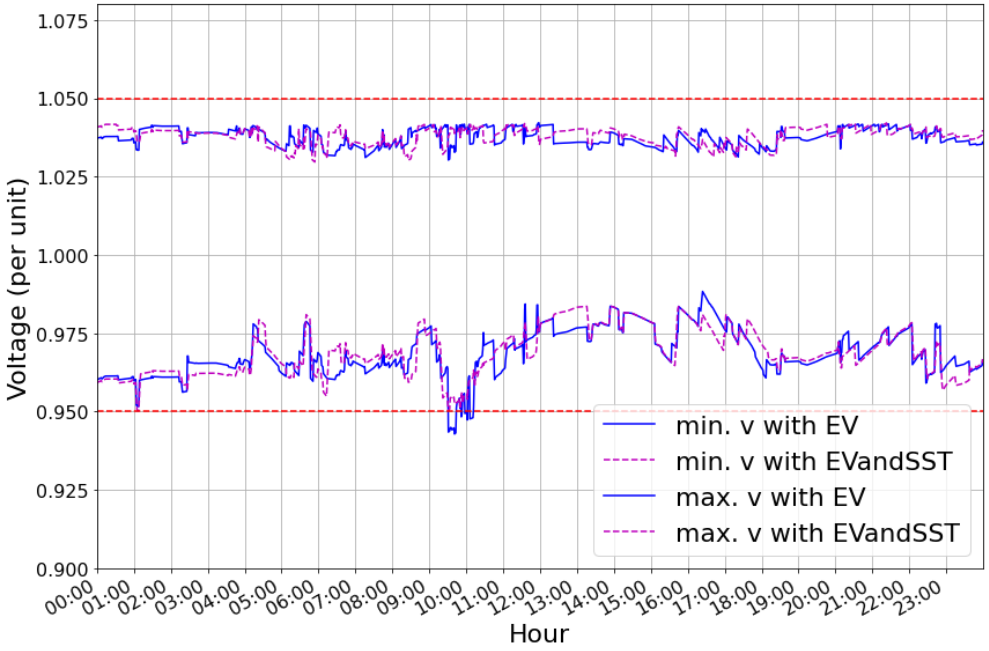


Figure 7. Overall minimum and maximum per unit voltage of the feeder in a day with and without SST in presence of high EV charging.

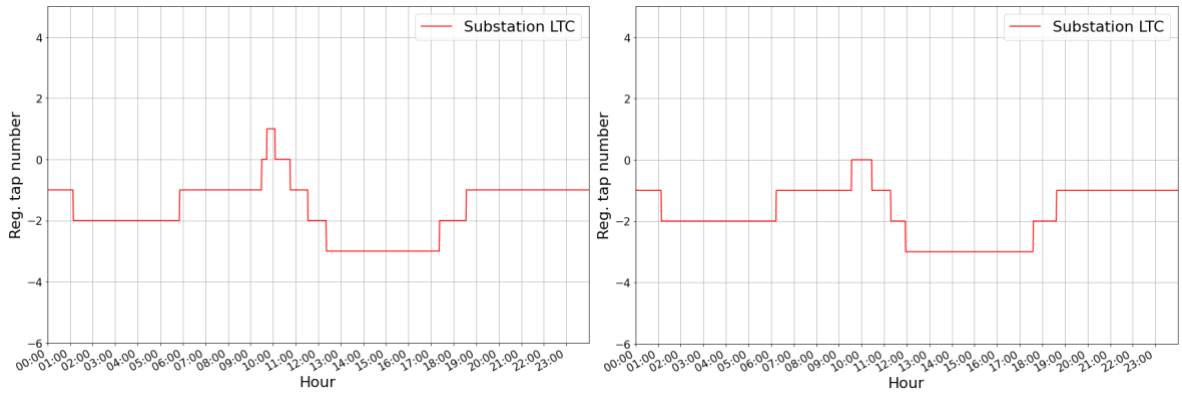
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Table 3. Voltage metrics

	Min per unit voltage	Max per unit voltage	No. of nodes below 0.95 p.u.
Scenario 1: With EV only	0.9428	1.0422	10
Scenario 2: With EV and SST	0.9502	1.0421	0

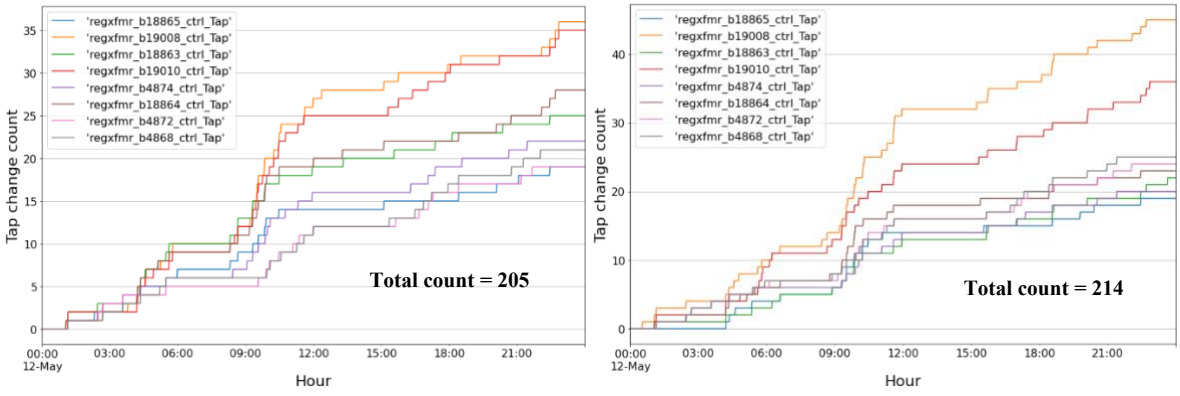
LTC and regulator taps:

The LTC tap position is shown in Fig. 8. The LTC tap position changed 10 times in the scenario with EV only, while it was reduced to 8 when SST was added. Fig. 9 presents the regulators tap change count, where the overall total tap change was observed to be 205 with EV only and it was increased to 214 when SST was added. However, in the scenario with SST, the voltage was better maintained.



(a) (b)

Figure 8. LTC tap (a) with EV only, (b) with EV and SST.



(a) (b)

Figure 9. Regulators tap change count (a) with EV only, (b) with EV and SST.

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SST power and voltage:

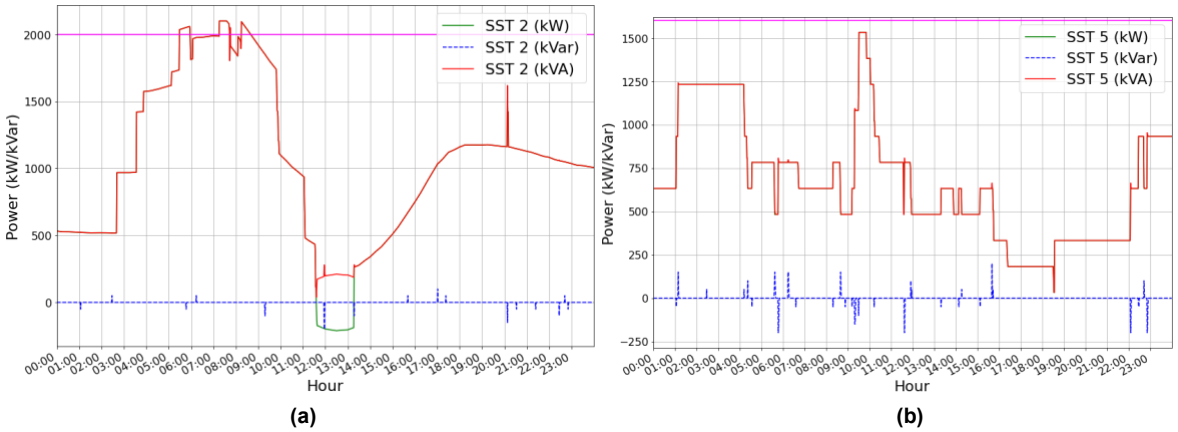


Figure 10. SST power.

Figure 10 portrays the active and reactive power of two of the SSTs. There is PV generation in the site B, Fig. 10 (a), because of which the active power (green curve) is negative during mid-day when the PV generation is high. Fig. 10 (b) shows the SST power at site E, which is at the farthest distance from the substation. The blue curve shows the reactive power of the SSTs, which means SST is consuming or generating reactive power at different instances. The MV and LV terminals per unit voltage profiles of the two SSTs are shown in Fig. 11. The LV terminal is maintained constant at 1 per unit when there is SST and the MV terminal voltages are regulated.

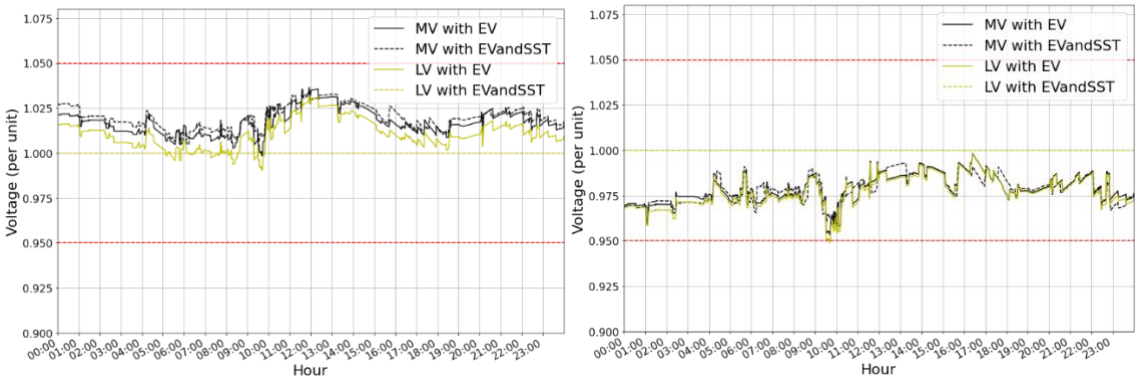


Figure 11. SST MV and LV nodes per unit voltages.

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Substation power:

The substation level active power slightly increased with SST while the reactive power reduced with SST as represented in figures 12 and 13 respectively.

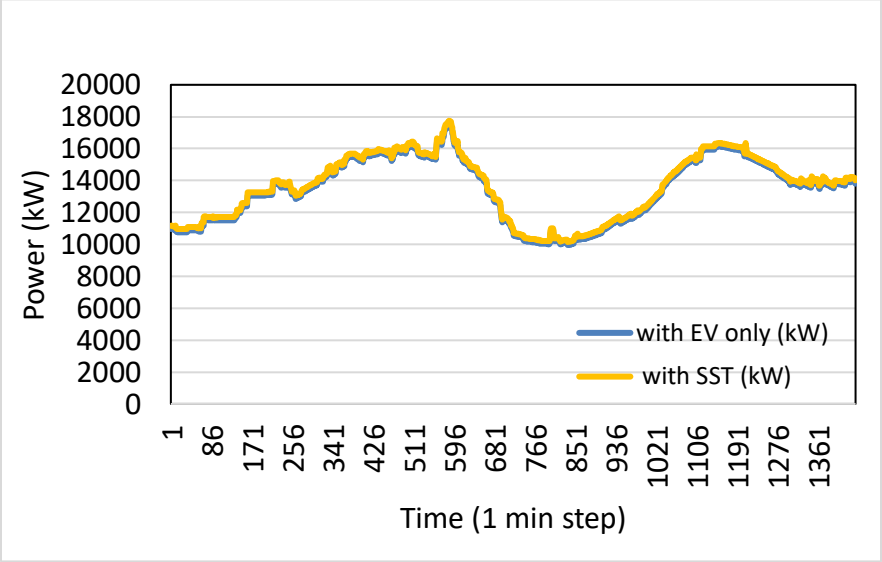


Figure 12. Substation active power.

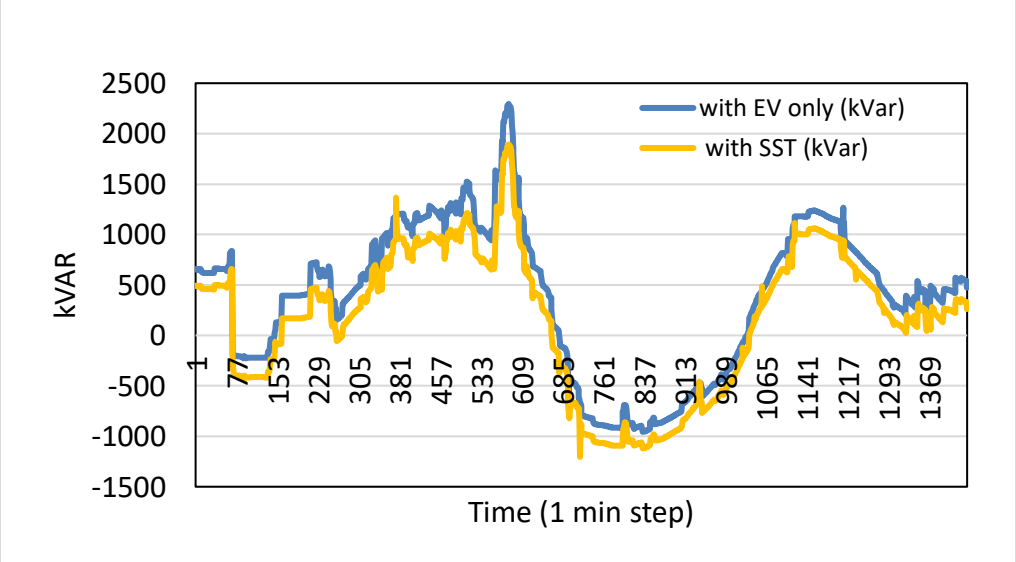


Figure 13. Substation reactive power

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Line loading:

Figure 14 shows the J1 network topology with line capacity loading at peak charging demand time with addition of EV charging at the sites. Overloading is observed in a stretch of lines before the site E, dark red section in the figure. So, this implies that site E may not be suitable location to consider for high EV charging in the existing J1 feeder.

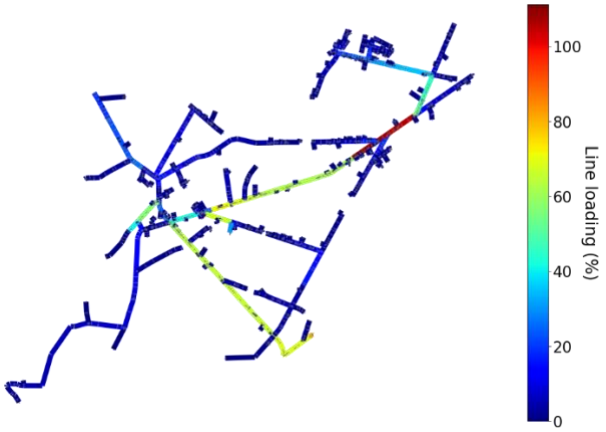


Figure 14. Line loading heat map at peak charging demand time.

Thus, the location of site E was updated to another location before the overloaded section in the feeder. The updated EV charging site is presented in Fig 15.

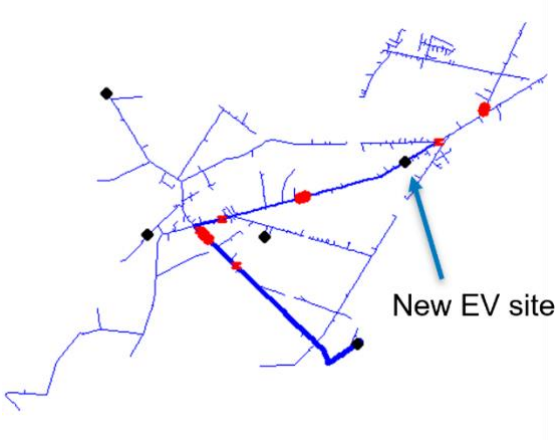


Figure 15. J1 feeder topology with updated EV charging site E, black marked points are the selected EV charging sites. Red markers represent the regulators and cap banks.

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Task 3 Identify potential customers and develop evaluation plan: The SST technology can facilitate the DER participation in the markets to provide grid services which would provide numerous sources of value to different stakeholders. The value could be in the form of improved grid control (volt/VAR control, storage), better visibility as a distributed sensor, synthetic inertia etc. Thus, the SST technology can benefit utilities, DER aggregators, balancing authorities etc. This task identifies the potential customers of the SST technology. An evaluation plan will be developed by NREL team to quantify the incremental value of the SST to these customers. The simulation scenarios will be designed by NREL team to extract the data required for this evaluation. Estimated completion: 9 months from project start.

The MV SSTs can provide benefits not only to the stakeholders such as utilities and DER developers/aggregators, but also to the balancing authorities and the customers. Some of the benefits to potential stakeholders and customers have been discussed here.

Benefits to stakeholders

1. Utilities

   SSTs can help the utilities with grid stability and reliability. SSTs can provide voltage and frequency regulation, and enable better integration of renewable energy sources, such as solar and wind, into the grid. SSTs can potentially reduce the need for maintenance and repairs, as they have fewer moving parts and are more durable than traditional transformers.

2. DER developers and aggregators

   SSTs can help DER developers and aggregators to better integrate their products into the grid. They can convert DC to AC and vice versa and contribute to voltage and frequency regulation. SSTs can help energy storage systems provide grid services, such as frequency regulation and peak shaving. This can help increase the reliability of renewable energy sources and make them more attractive to utilities.

   SSTs can help DER aggregators improve the performance of their systems, by providing faster response times and more precise control over power flow. This can help DER aggregators optimize the use of DERs, improve system reliability, and potentially increase revenue from ancillary services and demand response programs.

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SST benefits to Balancing Authorities

1. **Improved grid stability and reliability:** SSTs provide voltage and frequency regulation that improves grid stability and reliability. This is important for balancing authorities, as they are responsible for ensuring the real-time balance of supply and demand.

2. **Increased grid flexibility:** SSTs can enable the integration of renewable energy sources, such as solar and wind, into the grid. This can help balancing authorities meet their renewable energy targets and reduce their reliance on fossil fuels. SSTs can also help balance the grid during periods of high demand, by providing reactive power support and reducing the need for expensive peaking power plants.

3. **Reduced maintenance and repair costs:** SSTs have fewer moving parts and are more durable than traditional transformers. This can help reduce the need for maintenance and repairs, which can be costly for balancing authorities. Thus, SSTs can provide long-term cost savings despite higher initial costs.

4. **Improved efficiency:** SSTs are generally more efficient than traditional transformers, particularly when it comes to converting power from DC to AC or vice versa. This can help balancing authorities reduce their energy losses and improve overall grid efficiency.

Benefits to customers

SSTs can improve power quality for industrial and commercial customers, leading to increased efficiency and productivity. They can also help reduce downtime and maintenance costs, as they are less likely to fail than traditional transformers. SSTs can potentially lead to lower electricity costs for consumers, as they can improve grid efficiency and reduce the need for expensive upgrades to the grid infrastructure.

Task 4 Perform system level simulations and value analysis: In this task, the system level simulations are performed by NREL team for a range of use cases defined the previous tasks including “corridor charging” for EVs and solar plus storage (FERC 2222). The test data collected from the simulations will be used as an input to the SST value analysis. Visualizations are developed as needed to develop insights into the data. Estimated completion: 12 months from project start.

System-level Simulations

The defined use case scenarios in Task 2 were analyzed for 2 different conditions. One with fixed LTC, regulator taps, and cap controls disabled and the other with active LTC, regulator taps, and cap controls as presented in Table 4. The updated EV charging sites were considered, and the existing PVs were removed to focus on the SST impact on the grid. In the later phase, PV+storage are installed at the sites and the operation of SST with PV+storage was studied.

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Table 4. Conditions and use case scenario

Conditions	Scenarios within condition
Condition 1 **(Fixed LTC, regulator taps, and cap controls disabled)**	**1. Scenario with EV**
	2. Scenario with EV and MV SST
Condition 2 **(Active LTC, regulator taps, and cap controls)**	**1. Scenario with EV**
	2. Scenario with EV and MV SST

Additional scenario	
Scenario with DERs (Active LTC, regulator taps, and cap controls)	**Scenario with EV, SST, and PV + storage**

Result Analysis:

Condition 1 - Fixed LTC, regulator taps, and cap controls disabled

Fig. 16 shows the fixed positions of LTC, regulators, and cap banks. The simulations were conducted by keeping these devices at the shown positions for condition 1.

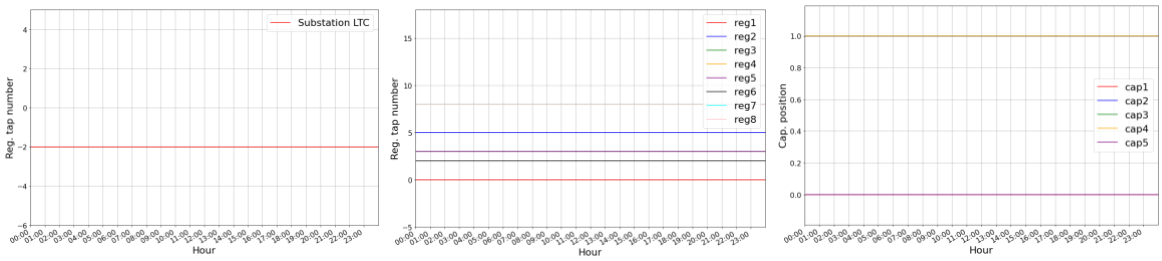


Figure 16. LTC, regulators and cap bank positions fixed.

Voltage:

The overall minimum and maximum voltages on the feeder through the simulation day are shown in Fig. 17, In scenario 1 with EV only, the minimum voltage goes beyond the ANSI limit of 0.95 per unit in the presence of high EV charging load. It's clear that the low voltage beyond 0.95 occurred whenever there is high coincidental EV charging. On the other hand, in scenario 2, when SST was added instead of conventional transformers, the minimum voltages were greatly improved. Even though the minimum voltages were raised significantly, the overall feeder minimum voltages were below 0.95 for high EV charging hours. This implies that the J1 feeder LTC, regulators and capacitors are required to be active for the feeder's daily operation.

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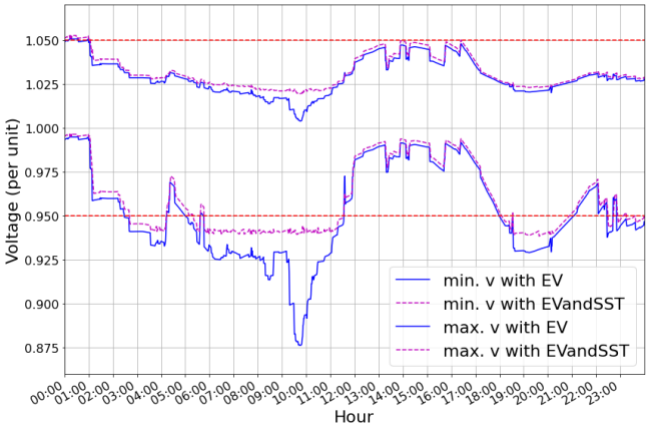


Figure 17. Overall minimum and maximum per unit voltage of the feeder in a day with and without MV SSTs in presence of high EV charging with fixed regulators and capacitors.

Regulators' voltages:

The fixed-tap regulators' primary and secondary voltages are presented in Fig. 18. Comparing the scenarios, the regulator voltages are at higher range than for the scenario with SST. It shows that SST is helping in uplifting the voltages.

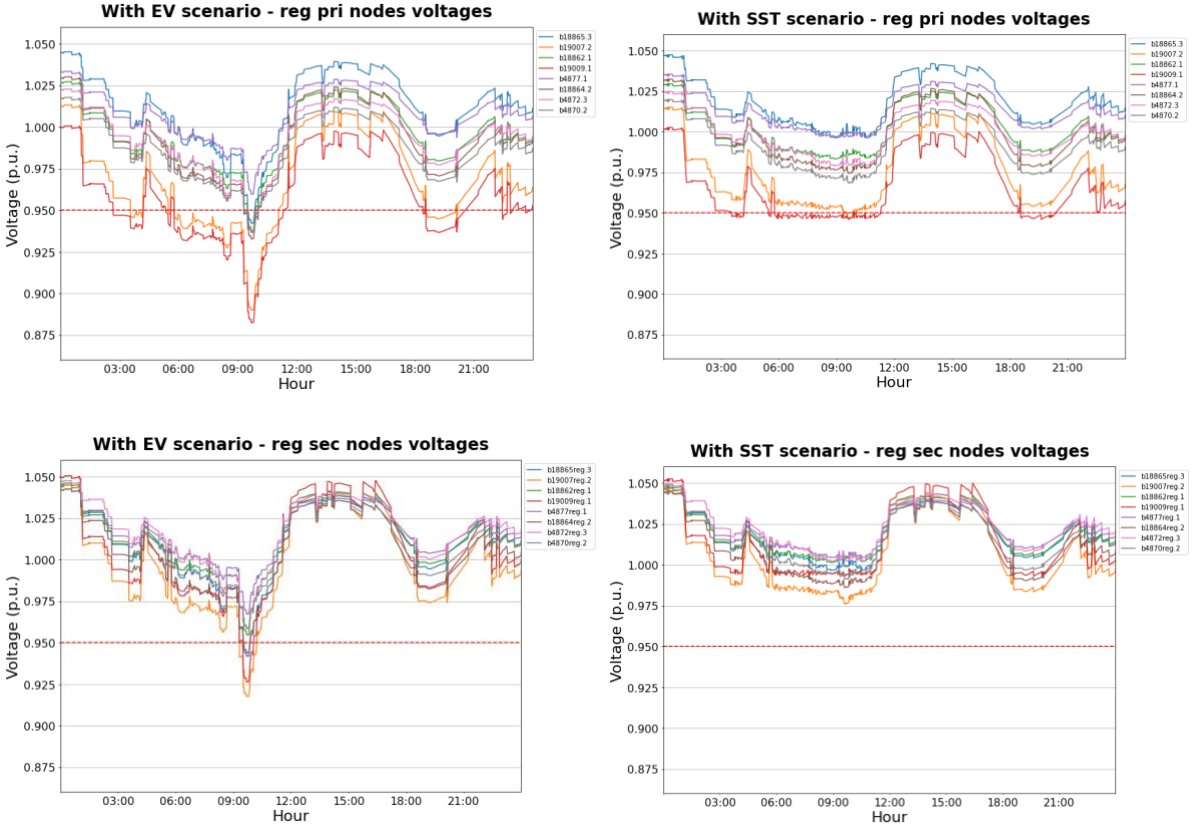


Figure 18. Regulators primary and secondary voltages through the simulation day.

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SST voltage and power:

The SSTs PCC voltages were maintained above 0.95 per unit as shown in Fig. 19(a). SST 2 and 5 had to provide reactive power support to maintain the voltage at different instances of the day. SST 5 had to provide continuous reactive power support during the peak charging demand hours, Fig. 19(b). Thus, the SSTs are actively regulating the site voltages as expected.

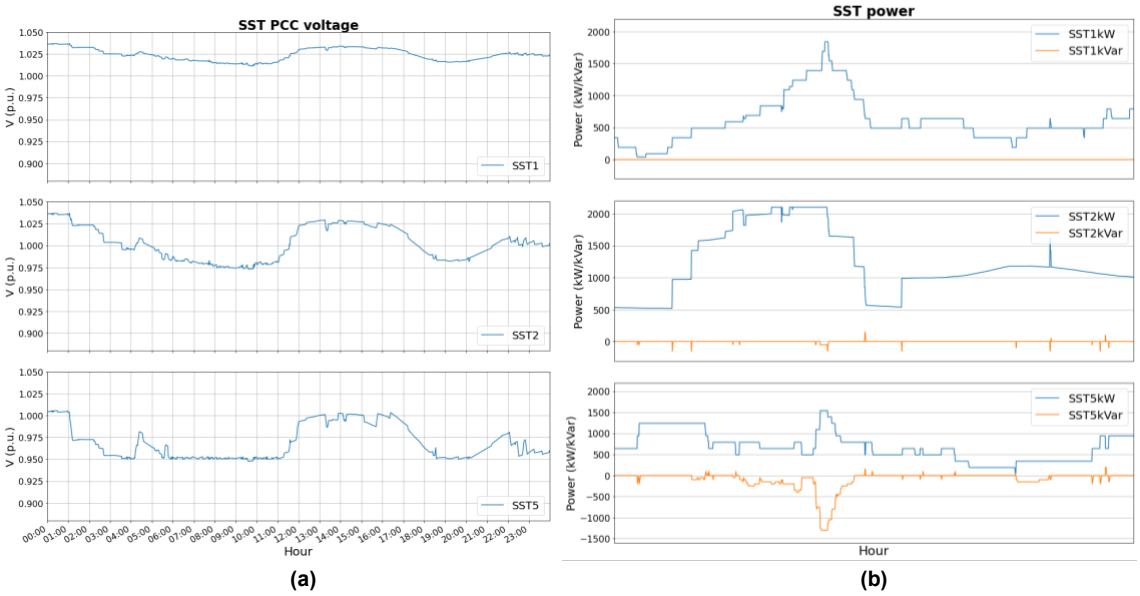


Figure 19. SST PCC voltage and power.

Fig. 20 (a) and (b) shows the MV and LV terminal voltages comparison for scenario with EV only and scenario with EV and SST. The black-dotted curve represents the regulated voltage at MV terminal with SST. Without SST the voltage at MV terminal had dropped below 0.95 per unit. Similarly, the SST also maintained the voltages at LV terminal at 1 per unit.

These results show that SSTs help to improve and maintain the local voltages, while this does not guarantee to regulate the overall feeder voltage. So, the LTC, regulators and capacitors are needed for J1 feeder.

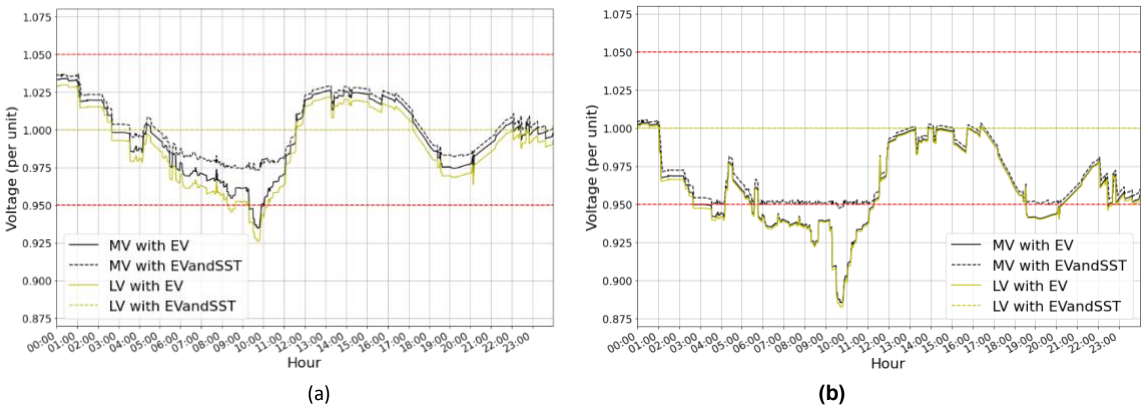


Figure 20. MV and LV terminal voltages without and with SST (a) SST 2, (b) SST 5.

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Condition 2 - Active LTC, regulator taps, and cap controls

Voltage:

The overall minimum and maximum voltages on the feeder through the simulation day are shown in Fig. 21. Without SST, the minimum voltage during feeder peak time dropped below the ANSI limit. The voltages are well-maintained in the scenario with SST. Thus, the SST can better support the overall J1 feeder operation in presence of the LTC, regulators and capacitors.

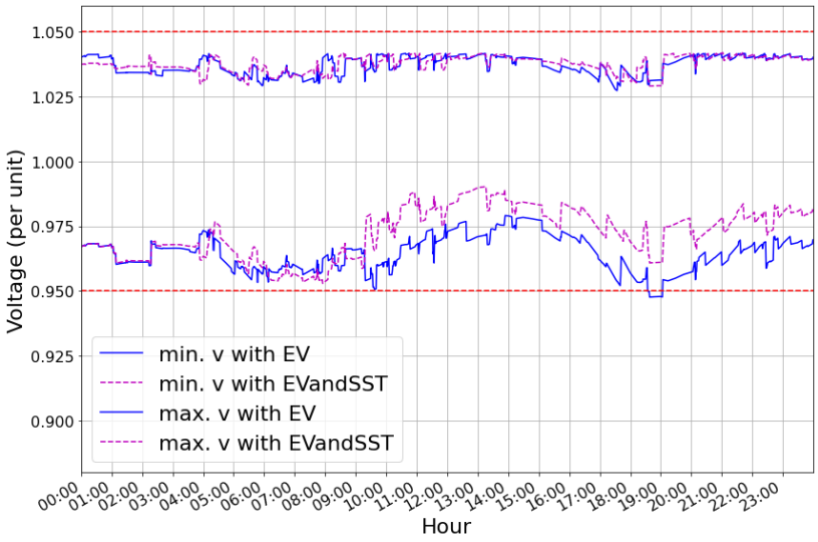


Figure 21. Overall minimum and maximum per unit voltage of the feeder in a day with and without MV SSTs in presence of high EV charging with active regulators and capacitors.

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Regulators' voltages and X/R ratio:

The regulators' primary and secondary node voltages with SST are shown in Fig. 22. The active regulators' taps maintain the voltages throughout the day as required by the feeder condition.

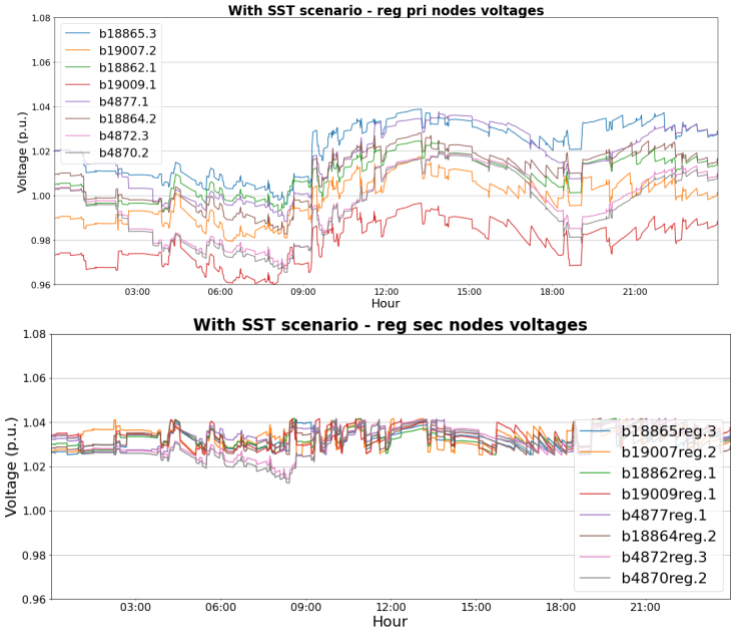


Figure 22. Regulators primary and secondary node voltages in presence of SSTs

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Fig. 23 presents the regulator taps and the X/R ratio at PCC node of SST 1 and SST 5. The X/R ratio is higher, closer to the feeder head (SST 1) and lower farther from the feeder head (SST 5). The change in X/R ratio at both SST nodes is reasonable. From the X/R ratios, it is observed that the change in load as well as the regulator tap positions influence its value.

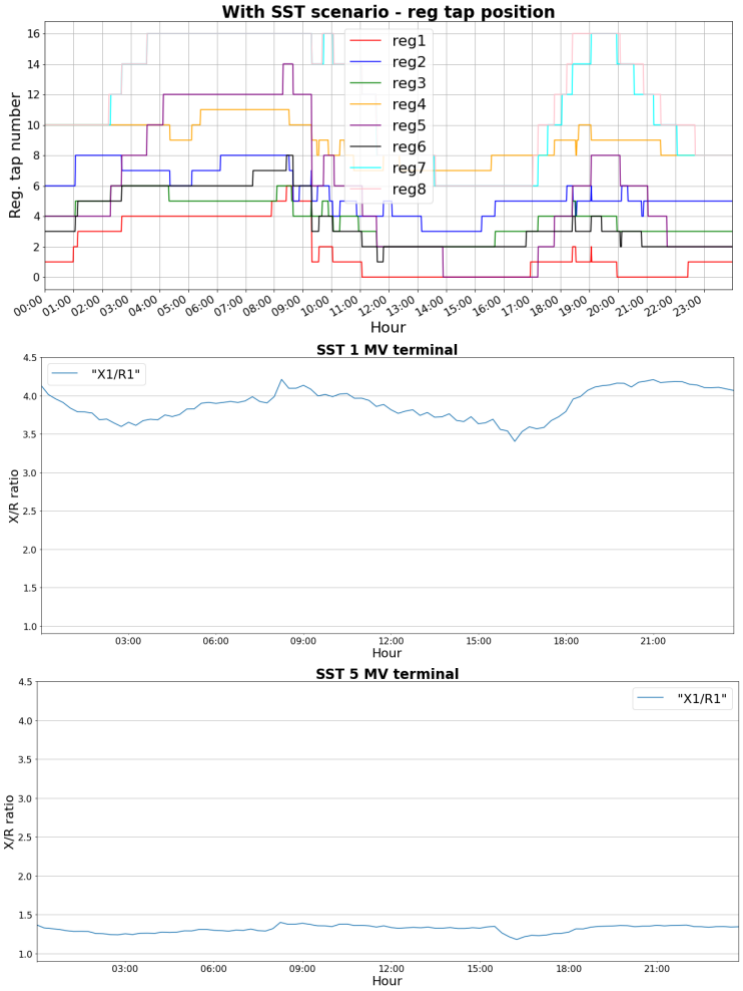
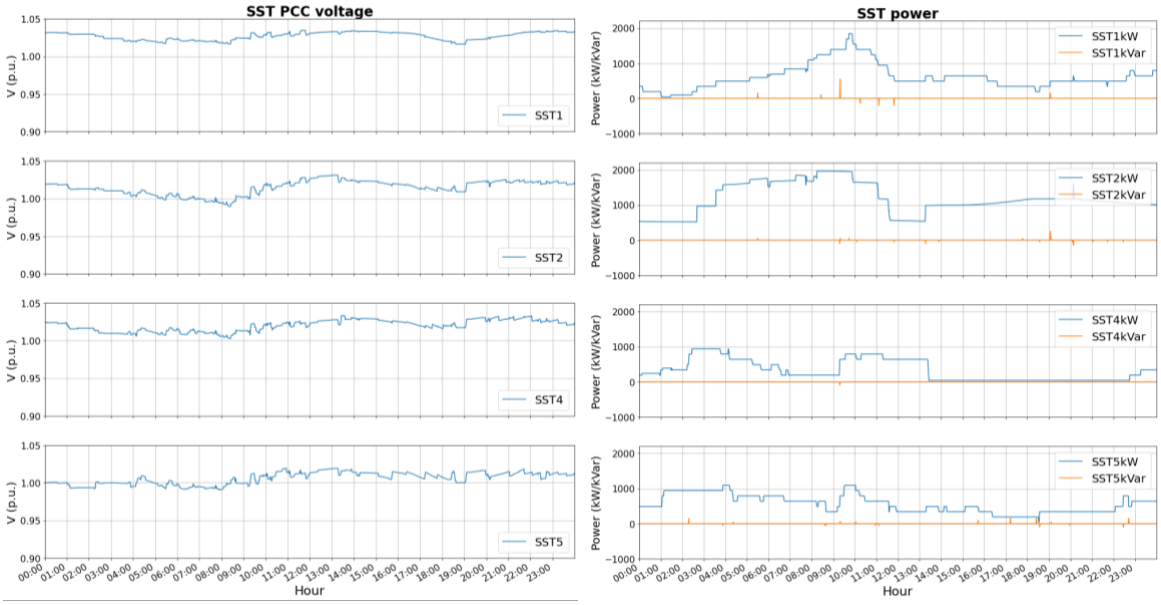


Figure 23. Regulators tap positions and X/R ratio at SST nodes.

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SST voltage and power:

The SSTs PCC voltages were maintained above 0.95 per unit as shown in Fig. 24(a). SST 2 reaches its max capacity during peak coincident charging while there is enough capacity remaining within other SSTs. Due to the active regulators and capacitors in the feeder, the SSTs MV terminal (PCC) voltages were within higher range and so the SSTs were not required to provide any significant reactive power support. There are only a few instances where the SSTs 1, 2 and 5 provided reactive power support.



(a) (b)

Figure 24. SST PCC voltage and power.

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Fig. 25 (a) and (b) shows the MV and LV terminal voltages comparison for scenario with EV only and scenario with EV and SST. The black-dotted curve represents the regulated voltage at MV terminal with SST. Similarly, the SST also maintained the voltages at LV terminal at 1 per unit.

These results show that SSTs help to improve and maintain the local voltages.

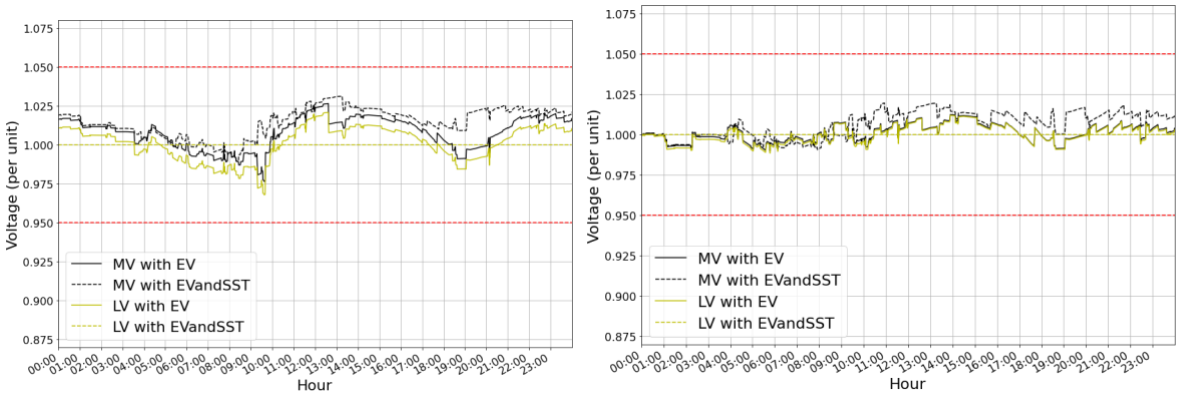


Figure 25. MV and LV terminal voltages without and with SST (a) SST 2, (b) SST 5

Additional scenario: Scenario with DERs

PV and energy storage were installed at EV charging sites. They were designed to reduce the feeder peak hour demand by providing feeder peak hours energy consumption for the site. They are designed to support the site peak charging demand as well. PV generation is utilized to charge energy storage. The J1 feeder peak hours extend from 4 to 9 pm. Table 5 shows the capacity of PV and energy storage considered. The charging loads were increased by adding 2 EV charging (added EVs scheduled to charge at peak charging time) to consider the increased capacity due to the DERs. Figure 26 shows the total EV demand before and after. The new demand is used in this analysis. The starting SOC of all the energy storages were 60% at the beginning of the simulation.

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Table 5. PV and energy storage sizing

PV + Storage size		
	PV size	**ES capacity**
Site A	340 kW	520 kW, 2492 kWh
Site B	928 kW	1420 kW, 6815 kWh
Site C	480 kW	735 kW, 3530 kWh
Site D	35 kW	50 kW, 240 kWh
Site E	225 kW	375 kW, 1640 kWh

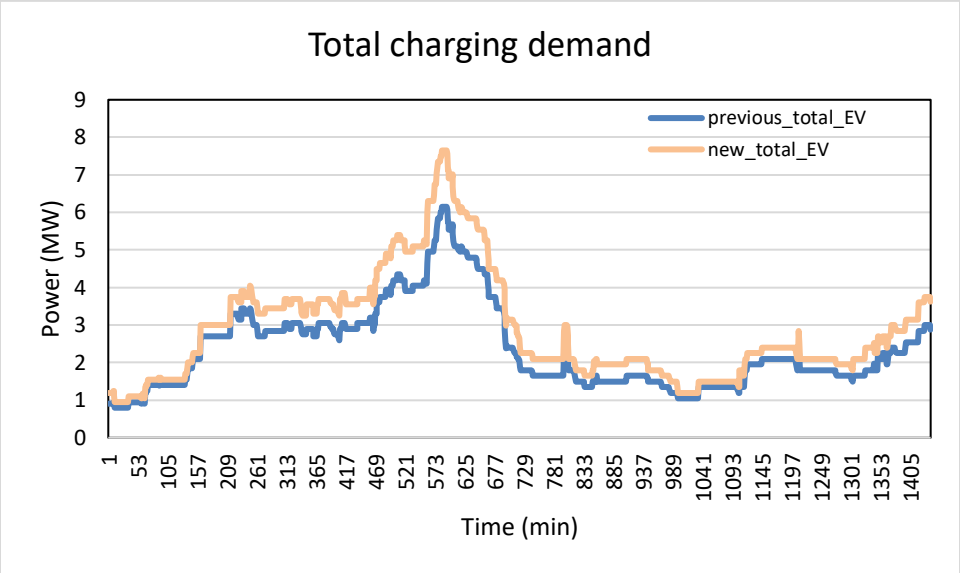


Figure 26. Total EV charging demand.

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Figure 27 shows the feeder voltage with DER scenario. Overall feeder voltage is regulated even with increased EV charging when there is SST and DERs on the sites. Likewise, Fig. 28 shows the SSTs power with the PV and energy storage implementation. It can be observed that the net site loads at the feeder peak window is minimized due to the DERs. It implies that SSTs support energy storage to provide peak shaving.

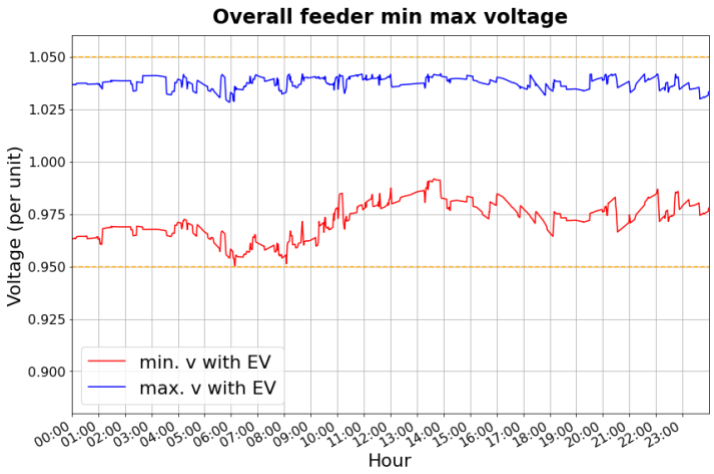


Figure 27. Overall feeder min and max voltage with DER and increased EV charging.

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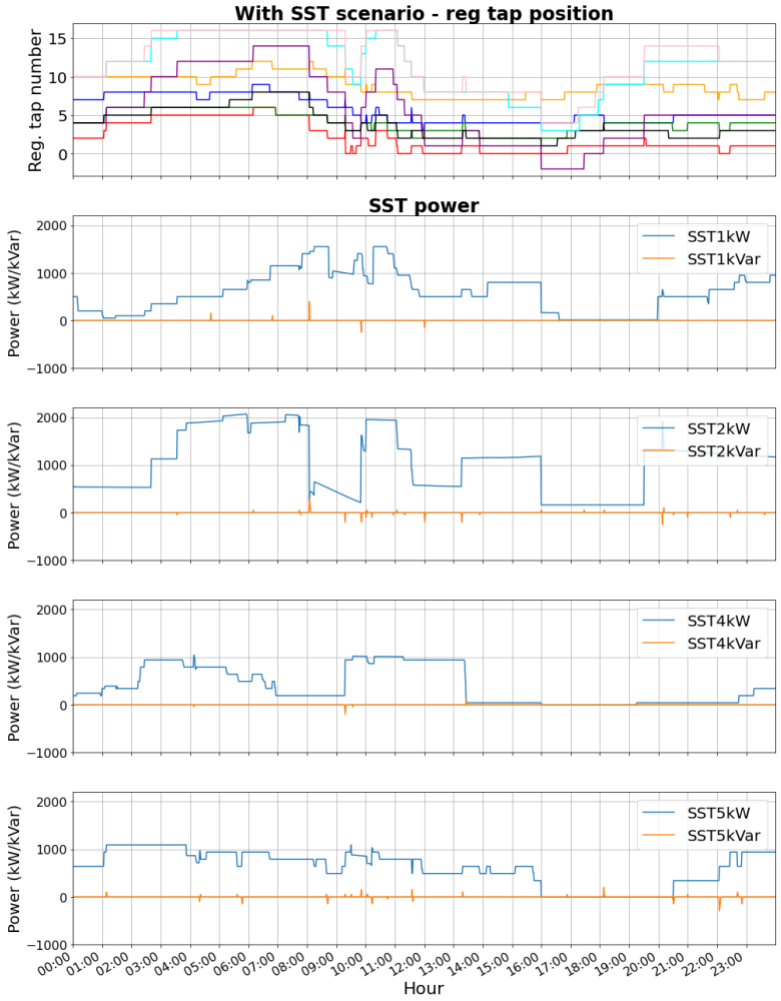


Figure 28. Regulators tap position and SST power with DERs installed at the sites.

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4.2 Value Analysis

An economic analysis was performed to explore the cost-benefit of a MV SST with DC-to-DC converters to provide Direct current fast charging (DCFC) charge regulation to multiple charge stations as against a conventional distribution transformer with downstream AC-to-DC rectifier and DC-to-DC converters. In the latter configuration, the low-frequency iron core transformer provides isolation, the DC-to-DC converter(s) controls the charge voltages, hence current is applied to the EV batteries according to the chosen charging strategy, and the AC-to-DC rectifier delivers an appropriate power level to the DC-to-DC converter(s). [2] The standard MV SST configuration of interest has an active AC-to-DC rectifier stage that feeds a DC-to-DC converter which can regulate DC voltage and supply power to a power inverter. If there is no AC secondary load, the inverter could potentially be eliminated. For a single charging station, the SST could be tapped at the regulated DC terminal to provide regulated DC power to an EV. To manage charging to multiple DCFC stations, however, a corresponding number of smaller DC-to- DC converters would be needed. Since either the rectifier or the DC-to-DC converters can provide isolation, it may be more cost-effective to provide isolation once in the rectifier, instead of in each converter, simplifying the converters. The economic opportunity to be evaluated consists of three parts. First, there are savings to be realized by substituting the SST for a conventional distribution transformer, an AC-to-DC converter, and multiple regulated DC outputs, obviating the cost of the conventional transformer in its own housing, its concrete pad, and its secondary connections and wiring to the AC-to-DC converter. Second, the expense of conventional transformer power losses will be avoided; the power electronics losses in the respective configurations will be roughly equivalent. Finally, there may be avoided costs through use of the MV SST's capability to provide reactive power onto the distribution system–a capability that is not typically available from behind-the-meter or secondary voltage power electronics in the configuration using the conventional transformer. This analysis considers the first two aspects of the economic opportunity, but not the last, as it is likely to be circuit-specific and depend upon circuit planning and design. This aspect does, however, have the potential to obviate capital expenditure for voltage regulation equipment and corresponding maintenance expenses. To be cost-effective, such an SST would have to have a lower life cycle cost than the conventional transformer, rectifier, converters, and conventional transformer losses. For the purposes of this rough cost-benefit analysis, we need not know the initial cost of the DCFC power electronics or an SST and can assume them to be roughly equal. Likewise, we can assume the losses in the SST and DCFC power electronics to be roughly equivalent, but for possible small additional losses in the SST for an isolating rectifier. If the avoided cost of the low frequency iron core transformer and its losses allows for a margin in the cost of the SST such that it can include nominal additional cost of an isolating rectifier, multiple DC-to-DC converters and marginally higher power electronics losses due to the isolating conversion, the SST in this application should be cost-effective. This is the cost comparison that was made in this analysis. The assumed economic perspective of the analysis is that of a taxable entity such as an investor-owned utility experiencing the cost-benefit of ownership and operation and maintenance of the distribution system. The analysis took the results of the simulated scenarios and calculated economic effects and compared the economics of the scenarios. The analysis also calculated projected maintenance expense due to switching operations of switching and tap-change devices on the system. The analysis assumed that the simulated 24-hour period was representative and typical and extrapolated the expenses from daily to annual, then to multi-year for discounted

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cashflow analysis. The numerical economic assumptions and the cost and maintenance assumptions made for the cost-benefit analysis are listed in Table 6.

Table 6. Assumptions for the Cost-Benefit Analysis

Economic numerical assumptions	
Cost of energy	$0.04
WACC	5.00%
Federal tax rate	21%
State tax rate	7%
Inflation rate	2.5% annual
Transformer/SST federal tax life	20 years
Transformer/SST state tax life	30 years
Cost and maintenance assumptions	
Number of switching cycles necessitating maintenance	100000
Regulator switch maintenance LTC maintenance	$60,000
Capacitor switch maintenance	$7,000
Transformer capital cost (conservative/likely)	172,000/248,000
Transformer annual maintenance cost	1.7% of capital cost

Results of Value Analysis

Table 7 shows 20-year cumulative net present values of costs (benefits) associated with the fixed switches scenarios. The break-even analysis, incorporating the economic assumptions as outlined above and the energy outputs of the simulations extrapolated out over a year, resulted in the SST capital cost set to $337,112.

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Table 7. 20-year net present values resulting from a MV SST with capital cost equal to the cost of DCFC power electronics plus $337,112 in the no switching scenario.

	Scenario	
Cost Class	**Legacy Fixed**	**SST Fixed**
Tap Maintenance	$0	$0
Regulator Maintenance	$0	$0
Capacitor Maintenance	$0	$0
SST Maintenance	$0	$64,697
Transformer Maintenance	$33,009	$0
Transformer Losses	$132,317	$0
SST Financing Payments	$0	$273,292
Transformer Financing Payments	$139,438	$0
Fed. Depreciation Credit	($28,317)	($55,500)
State Depreciation Credit	($6,239)	($12,333)
Total Cost	$270,155	$270,155

Because the MV SST cost in the SST scenarios and DCFC power electronics in the iron core transformer scenarios were assumed to be roughly equal, this indicates that the SST could cost $337,112 more than the DCFC power electronics and break even with a configuration with iron core transformer with secondary side DCFC power electronics. Table 8 shows 20-year cumulative net present values of costs (benefits) associated with the active switches scenarios with the SST capital cost set to $334,008 for purposes of break-even analysis. This indicates that in the active switching scenario, the SST could cost $334,008 more than the DCFC power electronics and break even with a configuration with iron core transformer with secondary side DCFC power electronics.

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Table 8. 20-year net present values resulting from a MV SST with capital cost equal to the cost of DCFC power electronics plus $334,008 in the active switching scenario.

	Scenario	
Cost Class	**Legacy Active**	**SST Active**
Tap Maintenance	$0	$0
Regulator Maintenance	$28,402	$29,176
Capacitor Maintenance	$0	$0
SST Maintenance	$0	$64,101
Transformer Maintenance	$33,009	$0
Transformer Losses	$130,604	$0
SST Financing Payments	$0	$270,775
Transformer Financing Payments	$139,438	$0
Fed. Depreciation Credit	($28,317)	($54,989)
State Depreciation Credit	($6,293)	($12,220)
Total Cost	$296,844	$296,844

Task 6: Grid-forming Inverter Support. Under this task, NREL and Eaton will support the Universal Interoperability for Grid-forming Inverter (UNIFI) consortium. This work will include hosting working groups on aspects of grid forming inverters modeling, controls, interoperability and developing specifications for grid forming inverters. The deliverable of this task will be a draft of specifications for grid-forming inverters. Estimated completion: 18 months from project start.

The purpose of this task was to support the Universal Interoperability for Grid-forming Inverter (UNIFI) consortium in developing a draft document of specifications for grid-forming inverters. This task planned and hosted working groups of national laboratory scientists, industry members, and academics to provide input on specifications. The draft specifications document was assembled and passed among the working groups for editing. A final draft was publicly posted on the UNIFI website with an open request for comments and feedback.

Summary

EPRI J1 feeder was selected for the distribution system analysis. The equivalent MV SST model for power flow was implemented and tested in OpenDSS and the reactive power functionality was developed. Five different sites were chosen as the high-power EV charging sites. The use case scenarios were developed to study the value of SST on the distribution feeder in presence of high EV charging. Using SST instead of conventional transformers showed the improved voltage profile, reduced substation reactive power demand, maintained the secondary side voltage at 1 per unit, and regulated the MV side voltage. Therefore, SST can benefit local sites by regulating voltage. It was observed that the regulators and capacitors installed at different locations in the feeder help maintain the overall feeder voltage, while having SST can benefit with improved voltages at conditions where regulators and capacitors were not able to maintain the voltage. The

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result analysis suggests that the integration of MV SSTs into the grid has the potential to enhance EV charging infrastructure, DER operation, and grid performance significantly. As the number of MV SSTs is increased, it is reasonable to expect even more substantial benefits for the overall grid stability and efficiency. This research paves the way for further exploration and deployment of MV SSTs in power distribution systems to accommodate the increasing demand for EV charging while ensuring optimal grid operation. Furthermore, the economic analysis indicates that an MV DCFC-configured SST could much more economically serve DCFC loads than T-R-D configurations. This represents a potential opportunity for electric utilities to provide DCFC service more cost-effectively through economic application of MV DCFC-configured SSTs. Additionally, further investigation into the coordinated operation of MV SSTs and the existing voltage regulators on the grid can be beneficial for the optimal operation of both devices.

Task 7 CRADA final report: NREL and Eaton will prepare a final report summarizing the results from this project. Preparation and submission in accordance with Article X. Estimated completion: 18 months from project start.

This report serves to meet the requirement for the CRADA Final Report with preparation and submission in accordance with the agreement's Article X.

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Subject Inventions Listing: None

ROI#: None

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